

Statistics Roadmap

6-Week Plan for Machine Learning → Deep Learning → NLP → Generative AI

Study time: 1.5 hours/day **Duration:** 6 weeks **Total:** ~63 hours

Week	Main Focus	Approx. Hours
1	Descriptive Statistics + Distributions	10.5
2	Probability + Bayes Theorem	10.5
3	Random Variables + Probability Distributions	10.5
4	Sampling + CLT + Inference + Hypothesis Testing	10.5
5	Correlation + Regression + MLE/MAP	10.5
6	Bias-Variance + Evaluation + Information Theory + GenAI	10.5

Daily 90-minute structure: 30 min concept/theory + 30 min Python/NumPy practice + 20 min ML/DL/NLP/GenAI application + 10 min revision/interview questions.

Week 1 — Descriptive Statistics

- Day 1: Population vs sample; types of data; descriptive statistics overview
- Day 2: Mean, median, mode; when each is useful
- Day 3: Range, variance, standard deviation
- Day 4: Percentiles, quartiles, IQR, five-number summary
- Day 5: Outliers; box plots; z-scores
- Day 6: Skewness, histograms, distribution shape
- Day 7: Python revision with NumPy/Pandas + mini dataset analysis

Why it matters: ML connection: feature understanding, outlier detection, scaling and exploratory data analysis.

Week 2 — Probability

- Day 8: Probability basics, sample space and events
- Day 9: Addition/multiplication rules; independent vs dependent events
- Day 10: Conditional probability
- Day 11: Bayes theorem
- Day 12: Random variables and expected value
- Day 13: Variance and covariance from a probability perspective
- Day 14: Python probability exercises + Naive Bayes intuition

Why it matters: ML/GenAI connection: classification, uncertainty and conditional next-token probabilities.

Week 3 — Probability Distributions

- Day 15: PMF, PDF and CDF

- Day 16: Bernoulli and Binomial distributions
- Day 17: Normal/Gaussian distribution and standardization
- Day 18: Uniform and Exponential distributions
- Day 19: Poisson and Multinomial distributions
- Day 20: Sampling from distributions with Python
- Day 21: Distribution revision + ML examples

Why it matters: ML connection: classification probabilities, modeling assumptions, noise and sampling.

Week 4 — Sampling & Statistical Inference

- Day 22: Sampling methods and sampling bias
- Day 23: Sampling distribution and Standard Error
- Day 24: Central Limit Theorem
- Day 25: Confidence intervals
- Day 26: Hypothesis testing; null vs alternative hypothesis
- Day 27: p-value, Type I/II errors and statistical power
- Day 28: Practical hypothesis-testing exercise + revision

Why it matters: ML connection: experiments, model comparison, A/B testing and uncertainty.

Week 5 — Relationships, Regression & Estimation

- Day 29: Covariance and Pearson correlation
- Day 30: Spearman correlation; correlation vs causation
- Day 31: Simple linear regression
- Day 32: Multiple linear regression; residuals
- Day 33: MAE, MSE, RMSE, R^2 and adjusted R^2
- Day 34: Maximum Likelihood Estimation (MLE)
- Day 35: MAP estimation + regularization intuition

Why it matters: ML connection: regression, parameter estimation, logistic regression and regularization.

Week 6 — ML + Deep Learning + NLP + GenAI Statistics

- Day 36: Bias, variance, underfitting and overfitting
- Day 37: Train/validation/test split and cross-validation
- Day 38: Classification metrics; confusion matrix, precision, recall, F1, ROC-AUC
- Day 39: Entropy and information gain
- Day 40: Cross-entropy and log-likelihood
- Day 41: KL divergence, mutual information and softmax
- Day 42: NLP/LLM sampling — temperature, top-k, top-p; final revision and interview questions

Why it matters: GenAI connection: language-model probabilities, token sampling, cross-entropy training and KL divergence.

Priority Checklist

Topic	Priority
Probability	★★★★★
Conditional Probability + Bayes	★★★★★
Probability Distributions	★★★★★
MLE / MAP	★★★★★
Bias-Variance	★★★★★
Cross-validation + Evaluation	★★★★★
Entropy / Cross-Entropy	★★★★★
KL Divergence	★★★★★
Correlation / Covariance	★★★★■
Sampling + CLT	★★★★■
Hypothesis Testing	★★★★■
PCA / advanced statistics	★★★██

Recommended learning method

- Learn the intuition before memorizing formulas.
- Implement important concepts using NumPy/Pandas.
- Immediately connect each concept to an ML/DL/NLP example.
- Maintain a one-page formula sheet and revise it every Sunday.
- After each week, solve 10–15 interview-style questions.

Target outcome: By the end of 6 weeks, you should have enough practical statistics to understand common ML/DL methods, NLP fundamentals, LLM training concepts and GenAI sampling—without studying statistics far beyond what these fields require.